Jackson Haiz
Prof. Groves
DS 5030
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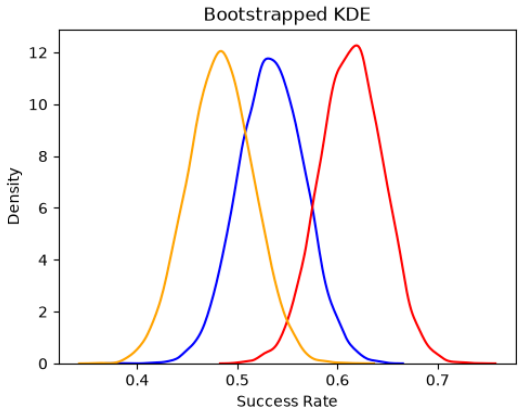
Lab 6 Write-Up

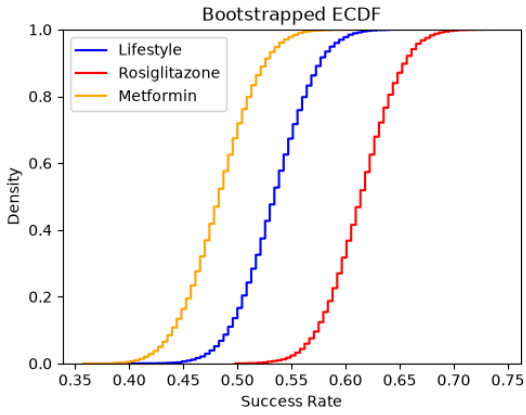
Analyzing the results from the three trials of treatments for pre-diabetics to reduce progression to full-on diabetes has yielded noteworthy results. Out of the three treatments, Lifestyle, Metformin, and Rosiglitazone, Rosiglitazone appears to be the most effective at accomplishing the goal of preventing progression to diabetes. I came to this conclusion through the use of multiple tests, each of which affirmed the notion resoundingly that Rosiglitazone is the most effective.

I first compared the outcomes of each of the three treatments and compared the rates of success across the three tests. The findings are below:

Treatment	**Rate of Success**
Lifestyle	53.4%
Metformin	48.3%
Rosigliatzone	61.4%

We can see from the table that the rate of success for Rosiglitazone is higher than the other treatments, with a 61.4% rate of success. Lifestyle treatments are the second most effective, with a 53.4% rate of success, and Metformin is the least effective, with a rate of Success of 48.3%. I then ran a bootstrap resampling model with 10000 different samples from the trials to stabilize the outcomes of each of the three tests. Doing this will not change the outcomes but will give us more stable results. With the Bootstrap results, I then plotted a Kernel Density Plot and an Expected Cumulative Distribution Function to illustrate the differences in outcomes across the three treatments. Attached below are the plots.





The plots back up the picture that has been painted through the outcome comparison. Rosigliazone has a more concentrated density at higher rates of success compared to the other two, which is reflected in its rightmost center in the Kernel Density Plot. The ECDF explains the same notion. Rosiglitazone is the rightmost distribution out of the three treatments, meaning that it has a higher success rate for each proportion of the sample. Despite Rosiglitazone being far and away the best treatment method for preventing progression to diabetes through statistical tests thus far, it is noteworthy to mention that Lifestyle and Metformin treatments are still effective options, just not to the degree of Rosiglitazone. Lifestyle appears to be the second-best option through the Bootstrap plots, being centered between Metformin and Rosiglitazone in the ECDF and KDE plots. Metformin is the least effective of the three, being centered to the left of the other two treatments in the two plots.

Now that we recognize which treatments appear better and worse than others through the trials, it is good practice to examine their Average Treatment Effects and plot them with confidence intervals. This explains how statistically significant one treatment's effectiveness is compared to another treatment, which makes a statement on the validity of choosing one treatment over another for the purpose of preventing diabetes progression. The average treatment effect is a comparison of one treatment to another treatment. The confidence interval measures the quantiles of the average treatment effect. For example, if Zero is in the confidence interval at the 99% confidence level, then there is not enough evidence at the 1% level to assume that the treatments are different. In other words, with 99% confidence, we cannot conclude that there is any difference in treatment effects. The null hypothesis is that there is no difference in effect of the two treatments. The Findings are below:

Bootstrap Confidence Intervals for ATE: Lifestyle - Metformin	**Bootstrap Confidence Intervals for ATE: Lifestyle - Rosiglitazone ***	**Bootstrap Confidence Intervals for ATE: Metformin - Rosiglitazone *** **
90% CI: [-0.0259, 0.1286]	90% CI: [-0.1565, -0.0025]	90% CI: [-0.2083, -0.0537]
95% CI: [-0.0428, 0.1454]	95% CI: [-0.1695, 0.0104]	95% CI: [-0.2212, -0.0406]
99% CI: [-0.0690, 0.1718]	99% CI: [-0.1994, 0.0403]	99% CI: [-0.2472, -0.0105]

While examining each confidence interval may be good practice, what we are particularly interested in is observing the difference in effect between the two highest-performing treatments, Lifestyle and Rosiglitazone. The confidence intervals tell us that, with 90% confidence, Rosiglitazone is superior to Lifestyle because the confidence interval's bounds are strictly

negative values. However, we cannot conclude that there is no difference in the effect of the two treatments with 95% or 99% confidence. This means that while there is noteworthy evidence to suggest that Rosiglitazone is the most effective treatment, we cannot be fully sure that it is more effective than the second-highest-performing lifestyle treatments because we could not conclude so at the 95% or 99% levels.

Ultimately, there is significant evidence to conclude that Rosiglitazone is the most effective treatment in preventing progression to diabetes. Between outcome comparisons, bootstrap kernel density functions, and bootstrap expected cumulative distribution functions, Rosiglitazone was consistently the most effective treatment option. However, when observing the confidence intervals for the average treatment effects across treatments, we failed to conclude that Rosiglitazone was superior in effectiveness to Lifestyle treatments with 95% or 99% confidence. Overall, analysis of these trials shows optimism for the treatments of preventing diabetes. As an analyst, I would love to look at more tests of such treatments. I believe that combining Data Science and Healthcare could raise welfare for patients and providers alike; I am proud to serve that purpose.